

Ms. Rashida**Web Engineering - CT-362**

Q.NO.1:

(Marks:12)[CLO-3:C4]

Analyze the requirements for a user registration form and write a JavaScript program that implements it with validation for the following fields:

- Username (must be 5-15 characters, alphanumeric with underscores and hyphens)
- Password (must be at least 8 characters with at least one uppercase letter, one lowercase letter, and one number)
- Email (must be in a valid email format)

On form submission, validate all fields using regular expressions and display appropriate error messages. Include both the HTML and JavaScript code.

Q.NO.2:

(Marks:12)[CLO-1:C2]

(A) Explain the purpose of the Document Object Model (DOM) and discuss the hierarchical relationship between the Window object, Document object, and Form object in the DOM.

(B) Discuss the stages of the ASP.NET Web Form page life cycle, from initial request to the final rendering of the webpage.

Q.NO.3:

(Marks:12)[CLO-3:C4]

Analyze and construct an ASP.NET solution to demonstrate how data is transferred between two web pages using the QueryString method by designing:

- Page1 with TextBox controls to input a user's name and email, along with a Button to submit the data.
- Page2 to retrieve and display the submitted information.

Q.NO.4:

(Marks:12)[CLO-1:C2]

Compare the following pairs of concepts by explaining how they differ in functionality and usage:

- a. DataReader and DataAdapter
- b. Persistent cookies and non-persistent cookies (with example)
- c. \$_GET and \$_POST superglobals (with example)
- d. cookies and sessions

Q.NO.5:

(Marks:12)[CLO-3:C4]

(A) Analyze the role of indexes in a MySQL database. Additionally, compare the different file modes available in PHP for opening a file.

(B) A customer relationship management (CRM) system requires functionality to both add new customers and update existing customer information in a database. Analyze the following requirements and write PHP code that:

1. Establishes a secure connection to a MySQL database named "myDB" using mysqli()
2. Performs two sequential database operations:

- o First, insert a new record with a user's first name, last name, and email into a table called "MyGuests"
- o Then, updates the last name to "Doe" for an existing record in the same table

3. Provides appropriate user feedback after each operation (success or error messages)
4. Properly closes the database connection when all operations are complete

Your solution should demonstrate proper handling of both INSERT and UPDATE SQL operations, including error handling for each operation independently.

SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (Computer Science)
SPRING SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2021

Dated : 24-JUL-24
Max Marks : 60

Web Engineering - CT-362

Q-01 CLO-01:

a) Scenario: Imagine you are tasked with developing a new mobile app that allows users to easily learn a new language. During the design phase, you identify a need for a speech recognition feature that grades a user's pronunciation. However, you discover this feature might require more development time than initially planned, potentially delaying the launch date. How would you approach this situation? Would you prioritize the launch date and potentially release a version without the speech recognition feature, or delay the launch to integrate the feature? Explain your reasoning and the potential pros and cons of each approach. [Marks 7.5]

b) Scenario: You are a software developer working on a new social media platform in the development phase. One of your teammates suggests a new and innovative design feature that could significantly improve user engagement. However, implementing this feature would require learning a new programming language by several team members. How would you evaluate and make a decision about this suggestion? Consider the potential benefits of the feature against the investment of time and resources needed to learn the new language. [Marks 7.5]

Q-02 CLO-03:

a) Develop a fully functional login page using PHP. The login page should consist of two input fields - one for the username and the other for the password - as well as a submit button. The only valid username should be 'Champion' and the corresponding password '123champion.com'. Upon incorrect input, an error message should be displayed in a label, and upon correct input, a success message should be displayed in a label. [Marks 10]

b) Explain how we must handle multiple incoming POST requests in PHP. Write an example code block to express your answer. [Marks 05]

Q-03 CLO-03:

a) Create a complete Cypress test file for the page in the question Q2(a). [Marks 06]

b) How does the Cypress test script validate the dropdown menu options and their values? Describe the process and the assertions used. [Marks 03]

c) Explain how the Cypress test script ensures that only one radio button can be selected. What are the key steps and assertions used in this test? [Marks 03]

P.T.O

- ✓ d) Suppose you must add a new input field for an email with validation that checks for a valid email format. How would you update the Cypress test script to include this new field and its validation? [Marks 03]

✓ Q-04 CLO-03:

Code the given interface using HTML, CSS & JAVASCRIPT with below mentioned validations: [Marks 15]

- ➔ First name and Last name must not be less than 3 characters.
- ➔ Username must have at least 5 letters followed by at least 3 digits.
- ➔ Email field input must be a valid email.
- ➔ Password field and Confirm Password field must have same input for successful form submission.
- ➔ An alert box must popup on invalid input as well as on successful submission

Register

Please complete to create your account.

First name

Last name

Username

Email

Password

Confirm Password

SIGN UP

[I am already a member](#)

SEAT NO. 32

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
THIRD YEAR (BS IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023

Batch 2020

Time : 3 Hours

Dated : 04-AUG-23

Max Marks : 60

Web Engineering - CT-362

Note: Attempt ALL Questions. All questions carry equal marks

Question 1: Explain how HTTP Request/Response works? Which codes are received in response and what does they mean? **[CLO-1, 12 marks]**

Question 2: Compare different types of APIs used to access database from web backend. Give an example of any two APIs using suitable code and comment which one do you think is better. **[CLO-2, 12 marks]**

Question 3: What are super globals? List down names of all super globals you know. Compare any two of them with suitable example / code. **[CLO-2, 12 marks]**

Question 4: Below is a form designed using web technologies. Implement form validation assuming that HTML and CSS work is done already. You can assume id, classname, field name etc. where and when required. **[CLO-3, 12 marks]**

Whoops! There were some problems with your input.

- The first name must be at least 5 characters.
- The last name field is required.
- The email must be a valid email address.
- The mobile no must be a number.
- The password field is required.
- The confirm password field is required.
- The details field is required.

First Name:

Har

The first name must be at least 5 characters.

Email:

asahid@nust.edu.pk

The email must be a valid email address.

Password:

Enter Password

The password field is required.

Details:

Enter Details

The details field is required.

Last Name:

Enter Last Name

The last name field is required.

Mobile No:

mobile

The mobile no must be a number.

Confirm Password:

Enter Confirm Password

The confirm password field is required.

Question 5: What are different methods for making a web application secure? Explain these methods along with suitable examples. **[CLO-1, 12 marks]**

BESE - 22

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR SPRING SEMESTER (SOFTWARE ENGINEERING)
EXAMINATIONS 2018
BATCH 2016-2017

Time: 3 Hours

Dated: 01-09-2018

Max. Marks: 60

Web Engineering - SE-206

Instructions:

- Attempt ALL questions.
- Make neat and clean diagrams where necessary.
- Questions can be attempted in any order but all parts of a question must be attempted together.

45 + 37

82

Q1	(a) There is a requirement to build web application "homeseconomy.com". The WebApp will serve as the hub for product marketing and sales, as well as the focus for a new generation of security monitoring services. You've been chartered with the job of building a Web Engineering team that will create the Web App. At the moment, nothing exists but a broad strategy and a few good ideas. The need is immediate, and a Web presence must be up and running as soon as possible. EXPLAIN how the Web Engineering framework activities will be conducted for the given scenario (1 iteration of WebE Process framework is required). [CLO-1]	5 [6]
(b)	EXPLAIN how a website can be secured using the HTTPS protocol. [CLO-1]	5 [6]
Q2	(a) EXPLAIN the Web Engineering (WebE) process framework with the help of diagram. [CLO-1]	6 [6]
(b)	DIFFERENTIATE between 3-tier and 2-tier Web Application Architecture with the help of diagrams. [CLO-1]	6 [6]
Q3	(a) Briefly EXAMINE how Security testing is done to validate the security of a web application. There is a popular news site. At any given time, 5000 concurrent users submit a request (a transaction T) once every 20 seconds on average. Each transaction requires the WebApp to download a news article that averages 15 Kbytes in length. Calculate the throughput and Link Speed for this web application. [CLO-2]	5 [6]
(b)	ANALYZE with the help of diagram the Database Testing mechanism for validating the Dynamic Contents of Web Application. Highlight the main layers of interaction as well. [CLO-2]	1 [6]
Q4	(a) A simple web form takes input from the user. If the number is less than 10 or greater than 30, an error message will be displayed. ANALYZE this input validation using Java Script Validation API. [CLO-2]	5 [6]

6

(b) There is a requirement to build a web application **mycompany.com.pk** comprising of different forms and pages. Following are the requirements [6]

- Form should have fields for Name, Comments, Email Address, Date
- Form should have an embedded CSS for presentation of font spacing, size and color.
- Input Validation should be done to minimize the load on the server.

ANALYZE this scenario to build a form fulfilling the requirements for this application using appropriate client-side technology. [CLO-2]

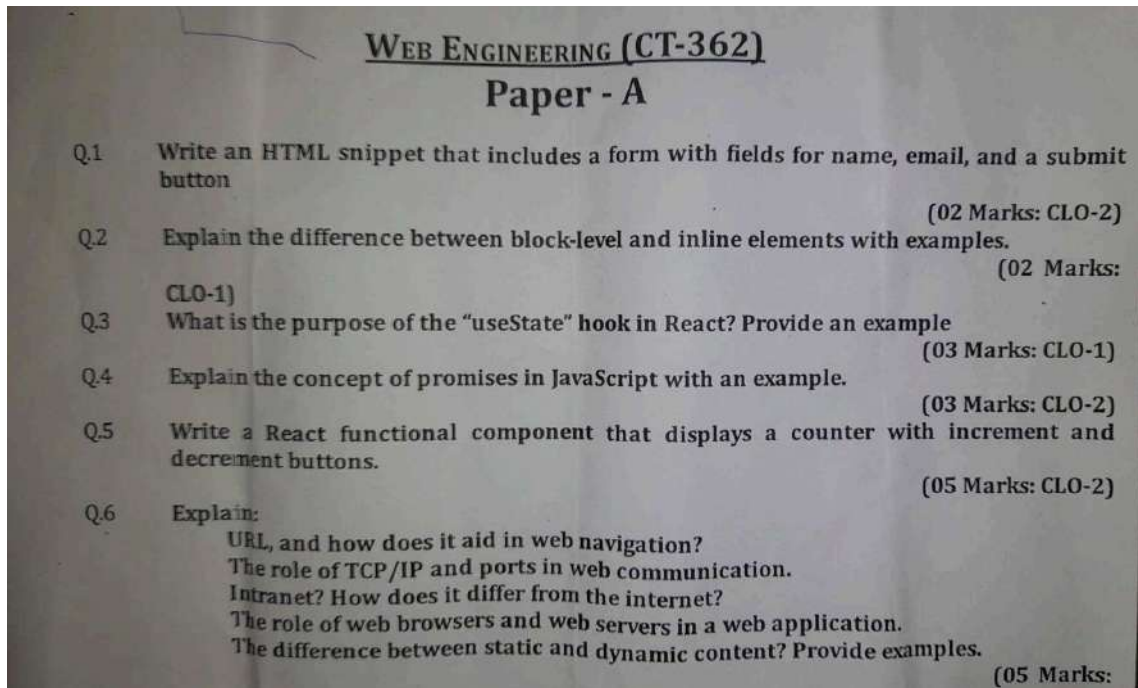
Q5

(a) Consider a web application **test.com** requiring backend connectivity. ANALYZE using either the MySQLi or PDO based object-oriented approach of PHP, how the following data can be inserted in database. Write down the relevant code. Also, do the input validation for email field. [CLO-2] 3 [6]

User id(primary Key)	First name	Last name	Date	Email
1	John	Mike	10/07/2018	john@test.com
2	Sam	Kim	11/07/2018	sam@test.com

(b) With the help of simple PHP form that takes three input from user "name", "password", "email" ANALYZE the difference between GET and POST PHP methods of Data submission. [CLO-2] 3 [6]

26/02/2025 Wednesday
Web Engineering
Sir Ayesha ur Rehman



Paper B:

Q1) HTML snippet that has navigation section with 3 links

Q2) block level Vs inline elements

Q3) purpose of useState and useEffect, example?

Q4) async, await in react

Q5) write a react functional component that fetches data from an API

Q6) same as Paper A